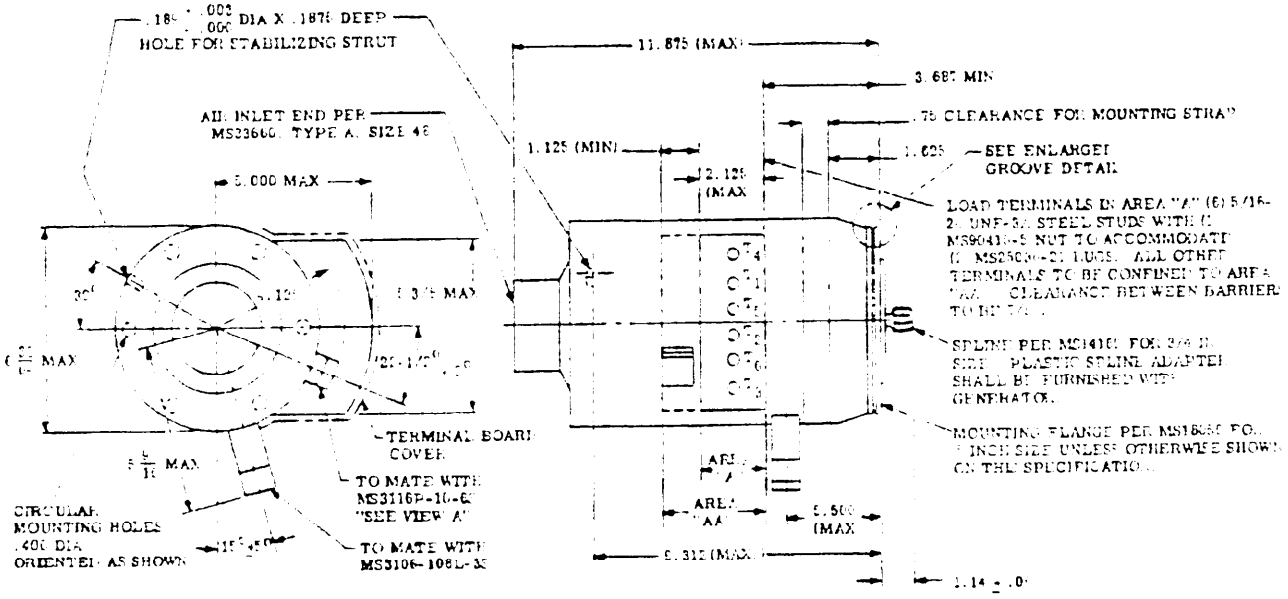
MILITARY SPECIFICATION SHEET

GENERATOR SYSTEM, 10 KVA, 400 HERTZ, ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS TYPE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator described herein shall consist of this document and the issue in effect of MIL-G-21480.



GENERATOR DATA	
RATED VOLTAGE	120/208
RATED OUTPUT	10 KVA
PHASE	3
RATED POWER FACTOR	0.75 LAG TO 1.0
FREQUENCY RANGE	380/420 HZ
SPEED RANGE	7600/8400 RPM
MAX SPEED FOR REGULATION	10000 RPM
OVERSPEED	11000 RPM
MIN EFFICIENCY AT RATED LOAD	80%
MAX WEIGHT	32 LBS
MAX OVERHUNG MOMENT	140 IN-LBS
SHEAR	690 IN-LBS
FLEXIBLE DRIVE	REQUIRED

FIGURE 1. M21480/11-1, GENERATOR.

FSC 6115

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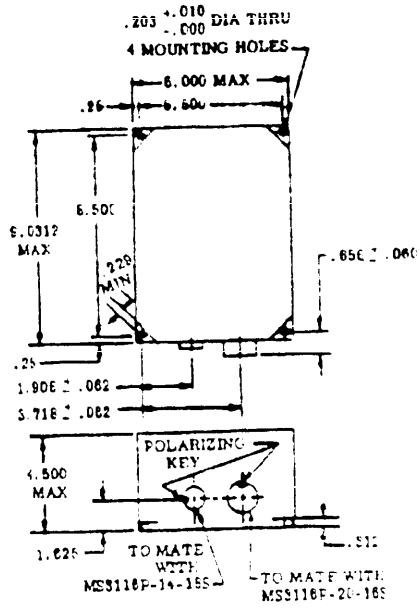


FIGURE 2. M21480/11-3 GENERATOR SUPERVISORY UNIT. MAX. WT. 3.5 LB

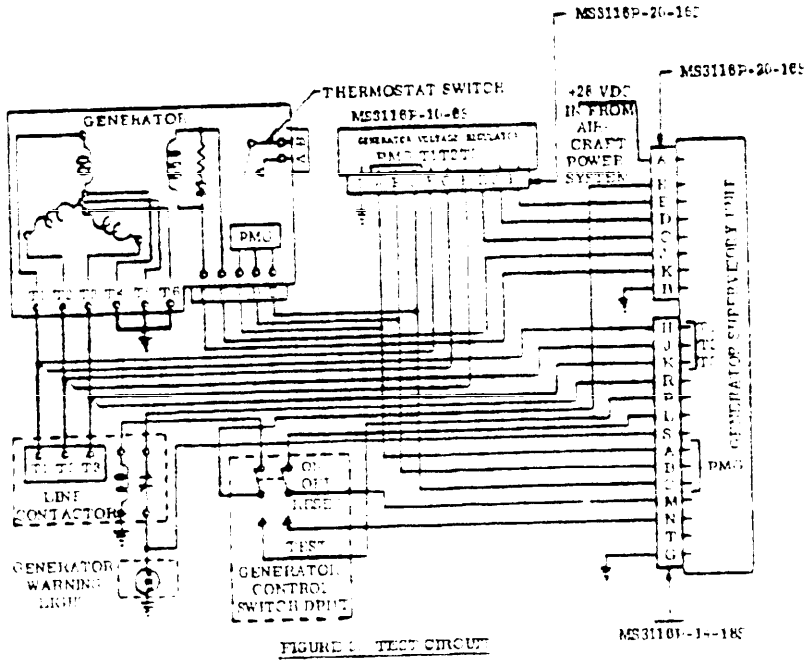


FIGURE 3. TEST CIRCUIT

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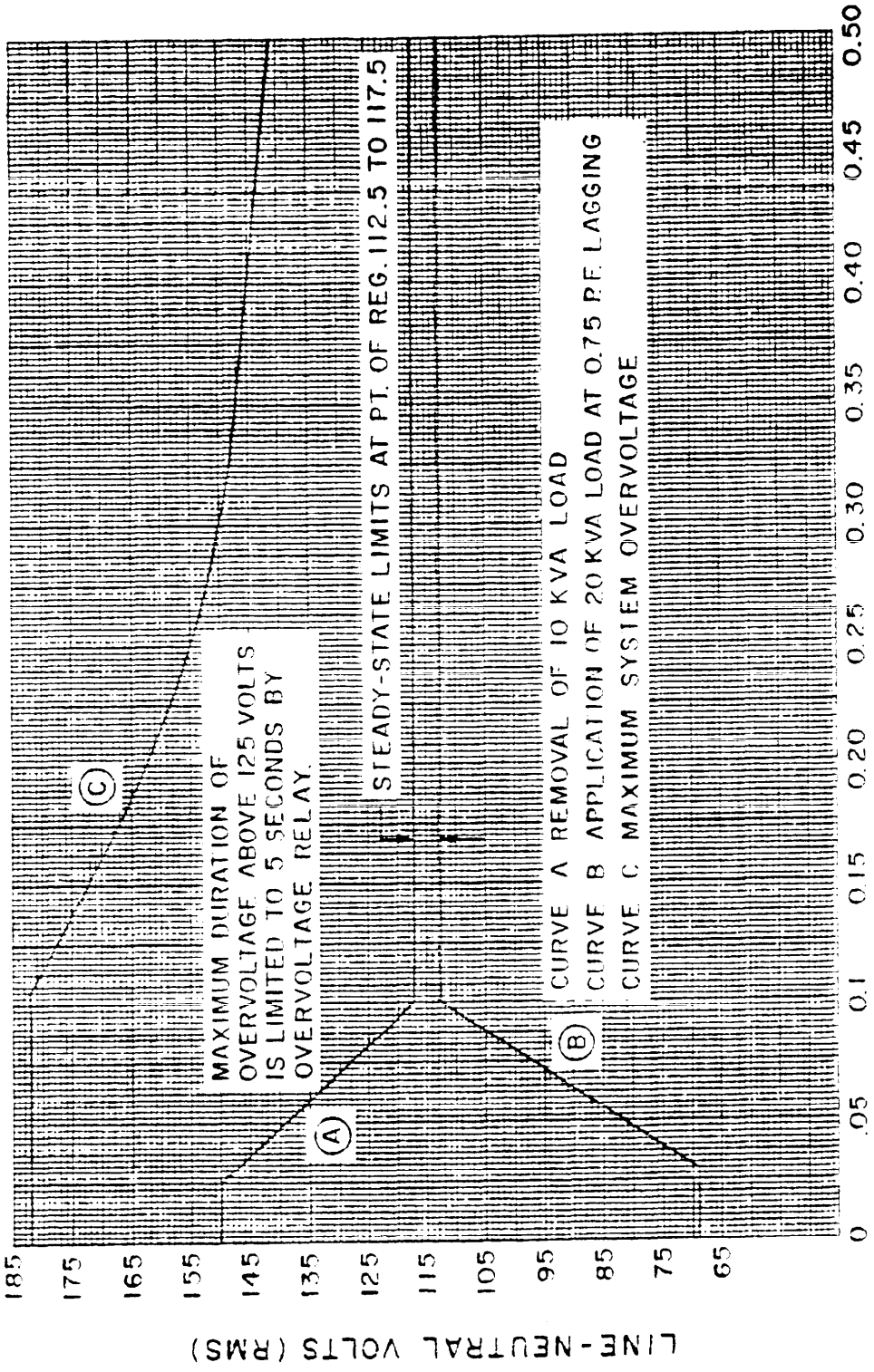


Figure 4. Transient, Steady State and Overvoltage Limits.

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REQUIREMENTS:

The requirements of MIL-G-21480 apply with the following exceptions to existing paragraphs:

Delete the following paragraphs:

3.4.2.2 through 3.4.2.8, 3.4.10, 3.4.13.1, 3.4.15.1, 3.4.16, 3.4.16.1 through 3.4.16.5, 3.5.7.6, 4.5.20, 4.5.22.1 through 4.5.22.3, 4.5.22.3.1 through 4.5.22.3.4, 4.5.24.1 through 4.5.24.3, 4.5.24.3.1 through 4.5.24.3.4, 4.5.25.1, 4.5.25.2, 4.5.25.2.1, 4.5.33, 4.5.33.1, 4.5.33.2.

3.4.2 Delete and add: Environmental Requirements. The system shall meet the requirements of MIL-E-81910 unless otherwise specified by the specification sheet.

3.4.2.1 Delete and add: Temperature and Altitude. The generator shall fully support the system when operated within the temperature-altitude requirements of MIL-E-81910 for blast cooled components. The control components shall operate continuously when subjected to the temperature-altitude requirements of MIL-E-81910 for self cooled components.

3.4.8 Change "Figure 3" to "Figure 4 of the specification sheet."

3.4.8.2 Change "5 minutes" to "2 minutes".

3.4.9.3 Add: Upon removal of any connector plug shown in Figure 3, the generator and generator voltage regulator shall be de-energized.

3.4.9.6.1 Add: An integral fan shall be incorporated to permit self cooling during operation under all electrical and environmental conditions specified herein.

3.4.9.6.1.1 Change "10,000 hour life" to "1000 hour life".

3.4.9.7 Delete and add: Life of Generator. The generator shall have a minimum of 1000 hours time prior to first overhaul.

3.4.10.3 Change "5000 hours" to "1000 hours".

3.4.13 Delete and add: Identification of Product. A nameplate conforming to MIL-P-15024, type A or B, permanently and legibly marked as required by MIL-STD-130 shall be securely attached to each system component. The nameplate shall contain at least the following information: Name of component, contractor's name or trademark, contractor's part number, contractor's serial number, federal stock number, and contract number.

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3.4.17 Delete and add: Reliability. The system MTBF (Mean Time Between Failures) shall not be less than 1000 hours.

3.5.1 Change "Figure 3" to "Figure 4 of the specification sheet."

3.5.4 Delete and add: Waveform. The crest factor for each phase voltage wave form shall be 1.41 ± 10%. No single harmonic shall exceed 3% of the fundamental.

3.5.5 Change "5 amps minimum continuous capacity" to "2 amps minimum continuous capacity".

3.5.7.2 Change "Figure 6" to "Figure 4 of the specification sheet."

3.5.7.3 Delete and add: Undervoltage. Undervoltage protection shall be provided by a static element which functions to disconnect the system from the load bus and deenergize the generator within a period of 6 ± 1 seconds after any phase voltage (line-neutral) falls to between 90 and 105 volts. The system shall be capable of being reset within 2 seconds when all three phase voltages are above drop out level and less than 105 volts.

3.5.7.4 Delete and add: Underfrequency. Underfrequency protection shall be provided by a static element which functions to disconnect the system from the load bus and deenergize the generator within a period of from 1 to 2 seconds after the frequency falls to 375 ± 5 Hz. The system shall be capable of being reset within 2 seconds when the frequency is greater than drop out level but less than 380 Hz.

3.5.8 Delete and add: Electromagnetic Interference. The system shall meet the requirements of MIL-E-81910.

3.7 Change "MIL-S-8879" to "MIL-S-7742".

4.2.1.1 Add: Qualification tests shall be supplemented with a minimum 100 hour failure-free flight test before consideration is given to incorporating the item on the Qualified Products List.

4.5.3.1 Change "Figure 2" to "Figure 4 of the specification sheet."

4.5.3.2 Change "Figure 2" to "Figure 4 of the specification sheet."

4.5.12 Delete and add: Electromagnetic Interference. The system shall be subjected to an Electromagnetic Interference Test in accordance with MIL-E-81910, except the ten microfarad feedthrough capacitor shall be removed.

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4.5.13 Delete and add: Wave form. The crest factor and harmonic content, line-line and line-neutral, of the output voltage shall be determined under the following conditions:

a. At generator speed of 8000 RPM for no load, and in 20% load increments to full rated load, unity power factor.

b. Condition (a) shall be repeated for 0.75 power factor.

c. Conditions (a) and (b) shall be repeated at generator speeds of 7600 and 8400 RPM.

The following requirements must be met during all above tests. The line-neutral crest factor shall be 1.41 ± 10%. No single harmonic shall exceed 3% of the fundamental.

4.5.20.1 Delete and add: Salt Fog, Control Components. Control components shall be subjected to a Salt Fog Test in accordance with MIL-E-81910, except the test shall be limited to 25 hours. At the completion of this test the unit shall successfully demonstrate its build-up characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.20.2 Delete and add: Oil-Salt Water Ingestion, Generators. The generator shall be subjected to an Oil-Salt Water Ingestion Test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its build-up characteristics and the effect of application and removal of rated and 200% load.

4.5.21 Delete and add: Fungus Resistance. System components shall be subjected to a Fungus Test in accordance with MIL-E-81910, with the installation instructions and mounting nuts attached. At the completion of this test, the unit shall successfully demonstrate its build-up characteristics, the effect of application and removal of rated and 200% load, and protective functions. Mounting nuts are to be unaffected and the package holding the nuts suitable for shipment. Instructions are to be legible.

4.5.22 Delete and add: Sand and Dust. System components shall be subjected to a Sand and Dust Test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its build-up characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.23.1.1 Delete and add: Generators. The generator shall be subjected to a Vibration Test in accordance with MIL-E-81910, except that the amplitude of vibration monitored at the anti-drive end of the generator during the main bending mode shall be limited to 20 g's but the vibration input shall not be lowered below 5 g's. At the completion of this test the unit shall successfully demonstrate its build-up characteristics and the effect of application and removal of rated and 200% load.

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4.5.23.1.2 Delete and add: Control Components. Control components shall be subjected to a Vibration Test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its build-up characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.24 Delete and add: Shock. System components shall be subjected to a Shock Test in accordance with MIL-E-81910. At the completion of procedure I, the unit shall successfully demonstrate its build-up characteristics, the effect of application and removal of rated and 200% load, and protective functions. There shall be no failure of the mounting attachments and the test item shall remain in place and not create a hazard during procedure III.

4.5.25 Delete and add: Humidity. System components, except those tested under paragraph 4.5.20.2, shall be subjected to a Humidity Test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its build-up characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.30 Delete first sentence and add: All system components except the generator shall be subjected to the Acceleration Test of MIL-E-81910. At the completion of the tests the unit shall successfully demonstrate its build-up characteristics, the effect of application and removal of rated and 200% load, and protective functions.

NOTES

1. Dimensions in inches. Unless otherwise specified, tolerances: linear 0.031, angular $\pm$ 1°.

2. Components shown enclosed in dotted lines not furnished under this specification.

ITEM	PART NUMBER
Complete System (Consisting of one Generator, one Generator Voltage Regulator, and one Generator Supervisory Unit)	M21480/11-10
Generator	M21480/11-1
Generator Voltage Regulator	MS25366-2
Generator Supervisory Unit	M21480/11-3

For design feature purposes, this specification takes precedence over procurement documents referenced herein. Referenced documents shall be of the issue in effect on date of invitations for bid.

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